Midterm Learning Evidence

AGAP-Link: Community Emergency Reporting and Incident Dispatch System

In Partial Fulfillment of the Requirements for the Course

ITC104

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INTRODUCTION

Introduction

Emergency situations such as structural fires, medical distress, and localized security threats require timely communication and coordinated response. During an emergency, the ability to obtain accurate information about an incident and its location is vital because responders need sufficient data to determine the appropriate course of action and allocate required resources. In the Philippines, communities frequently confront critical public safety hazards that demand rapid local intervention. The World Bank (2023) emphasizes the importance of strengthening disaster risk management and operational resilience at both national and community levels. In high-density municipal settings such as Bacolod City, fulfilling this mandate requires responsive local infrastructure capable of coordinating emergency interventions across barangay and citywide jurisdictions.

The effectiveness of emergency response is strongly influenced by the availability and quality of operational information. During emergencies, information must not only be received quickly but must also be accurate, accessible, and structured for decisive action. Chan and Tsai (2023) explained that disaster-response organizations face increasing volumes of information from disparate sources, creating severe challenges in processing reports and maintaining situational awareness in a timely manner. Their study demonstrated the use of instant-messaging-based technology to improve the accessibility and timeliness of data for governmental response decision-making, indicating that digital communication platforms can substantially support emergency management by reducing delays in transmission and data handling.

Mobile technology has increasingly been explored to bridge the gap between affected citizens and emergency responders. Berawi et al. (2021) developed the SaveMyLife mobile application to enhance communication between disaster victims and rescuers, focusing on prioritizing actionable information for rescue teams. Their findings concluded that mobile applications effectively support communication during crisis events across both preparedness and active response phases. Similarly, Nasir et al. (2022) developed and evaluated an Android-based mobile panic button application as an emergency notification service, demonstrating through user and officer simulations that mobile platforms can expedite notification delivery and response coordination.

Although these implementations demonstrate the value of mobile and location-based reporting, citizen reporting constitutes only the front end of a complete emergency response lifecycle. An effective emergency system must simultaneously facilitate the transmission, triage, monitoring, and dispatch coordination of reported incidents with authorized personnel. Chan and Tsai (2023) noted that multi-agency involvement often complicates information exchange without a centralized platform. Furthermore, Ignaco (2021) incorporated responder coordination into an incident-reporting application by enabling responders to manage incidents and coordinate supplemental response units.

To address these local communication bottlenecks, this study proposes the design and development of AGAP-Link: Community Emergency Reporting and Incident Dispatch System for Bacolod City. Rather than replacing existing emergency hotlines or agency communication lines, AGAP-Link serves as an integrated digital platform comprising a citizen-facing mobile application and a centralized dispatch management component. The system enables residents to transmit structured incident reports—including incident classification, photographic evidence, timestamps, and automated GPS coordinates—directly to dispatchers. By integrating mobile reporting, automated geolocation, offline queuing, and centralized dispatch tracking into a single unified architecture, the system aims to reduce reporting delays, minimize verbal miscommunication, and support the development of a safer and more resilient community (Berawi et al., 2021; Ignaco, 2021; Nuevas et al., 2024).

Background of the Study

Rapid urbanization and increasing population densities present continuous challenges for local communities, particularly in maintaining public safety and executing fast, synchronized emergency interventions. Emergencies such as structural fires, medical distress, and security threats demand immediate action from local authorities and barangay responders. However, conventional public safety frameworks heavily rely on traditional communication channels, such as manual telephone hotlines and fragmented word-of-mouth networks. These legacy systems frequently introduce critical communication bottlenecks, including congested lines, delayed information dissemination, and a profound lack of real-time situational awareness. In many developing regions, these communication challenges severely hinder the coordination of emergency medical dispatch and rapid response efforts (Friesen et al., 2024). Consequently, these systemic communication gaps extend emergency response times, putting community safety and property at heightened risk during critical incidents.

To bridge these vulnerabilities, modern emergency management strategies increasingly leverage mobile technology and smartphone-based applications to transform how citizens report incidents. The integration of mobile applications equipped with Global Positioning System (GPS) capabilities allows for rapid location detection and alert notification, fundamentally improving how emergency cases are reported directly to authorities (Tortor et al., 2024). Furthermore, deploying localized, mobile-based GPS applications ensures that citizens can immediately connect with local dispatch units through a seamless reporting mechanism (Nuevas et al., 2024). By establishing a structured digital framework, vital emergency alerts and geographical data can bypass the limitations and inaccuracies of traditional, verbal-only emergency calls.

Effective community resilience requires a seamless connection between the reporting resident and the emergency dispatcher. Research demonstrates that an integrated system comprising a mobile application for the public and a centralized web dashboard for dispatchers can significantly reduce intervention times by transmitting precise text and multimedia data, which overcomes the panic, noise, and language barriers often experienced during frantic oral calls (Repanovici et al., 2022). Additionally, modern centralized response architectures utilize automated dashboards to collect, process, and disseminate incident data instantly to specific responder units, such as fire brigades, medical personnel, or police units, thereby minimizing human processing delays and improving dispatch coordination times (Mohsin & Muyeed, 2024). Aligning directly with Sustainable Development Goal (SDG) 11, these robust digital tools are essential to making cities more inclusive, safe, resilient, and sustainable.

In the Philippines, municipal-level digital adoption has gained significant traction through platforms such as the MyNaga App (City Government of Naga, 2025). Developed as an official digital gateway in partnership with the local government, MyNaga demonstrated the value of civic mobile applications by centralizing emergency hotlines and allowing residents to report community issues directly from their smartphones. While MyNaga provides a wide-ranging civic and public service portal, high-density urban areas like Bacolod City require a purpose-built, high-urgency system focused specifically on end-to-end emergency triage, live tracking, and responder dispatch.

Drawing inspiration from this civic-tech model and addressing localized operational challenges, this study introduces the AGAP-Link: Community Emergency Reporting and Incident Dispatch System for Bacolod City. Consisting of the AGAP-Link Mobile Application for residents and a centralized web dashboard for dispatchers, the system resolves the friction points of conventional reporting mechanisms. AGAP-Link empowers Bacolod residents to transmit critical incident data—including precise GPS coordinates and attached photo evidence—while equipping local dispatchers and response units with the centralized tracking and triage mechanisms necessary to elevate operational efficiency and safeguard community well-being.

Statement of the Problem

Emergency situations in Bacolod City require fast, accurate, and coordinated communication between residents, emergency dispatchers, and local responder units. However, traditional emergency reporting channels within the city continue to rely on voice telephone hotlines and fragmented communication methods. These channels frequently result in congested lines during peak crises, delayed information dissemination, incomplete incident descriptions, and a lack of real-time situational awareness. These deficiencies make it difficult for emergency responders to promptly determine the exact location, nature, and severity of an incident, leading to preventable response delays.

Although mobile technology and GPS-based reporting tools can improve the transmission of incident details, there remains an operational gap in connecting citizen-generated reports with a structured triage and dispatch workflow. Community members require a straightforward platform to submit verified emergency reports containing essential details such as incident categories, photographic evidence, timestamps, and automated geographical coordinates. Concurrently, authorized emergency personnel require a centralized dispatch mechanism to receive, assess, prioritize, and coordinate these incoming incident reports in real time.

Therefore, this study seeks to determine how a community emergency reporting and incident dispatch system can improve the speed, accuracy, accessibility, and coordination of emergency response in Bacolod City. Specifically, this study aims to address the following questions:

1. How can the mobile application facilitate fast, structured emergency reporting by residents using one-tap category selection, automated GPS coordinate capture, and photographic evidence?
2. How can the system support continuous reporting capability when active cellular data or internet connectivity is unavailable?
3. How can the system provide real-time dispatch tracking and management, allowing reporting residents to monitor the status of their reports while enabling authorized dispatchers to receive, review, and update response statuses?
4. How can the mobile application provide residents with immediate, offline-accessible contact information for local emergency responder units and barangay offices?

System Objectives

The general objective of this study is to design and develop the AGAP-Link: Community Emergency Reporting and Incident Dispatch System, an integrated platform that enables Bacolod City residents to report emergencies quickly through an Android mobile application and empowers authorized personnel to receive, triage, and track incident dispatches through a centralized web dashboard.

Specifically, the study aims to achieve the following objectives:

1. **Enable Fast and Structured Emergency Reporting:** Develop a mobile reporting module within the AGAP-Link Mobile Application that allows residents to report incidents (specifically structural fires, medical distress, and security threats) using one-tap category selection, automatic mobile GPS location detection, and photo evidence attachment.
2. **Support Resilient Offline Report Queuing:** Implement a localized offline logging mechanism that stores emergency reports securely on the device when internet connectivity is unavailable and automatically synchronizes and transmits the queued reports once network connectivity is restored.
3. **Facilitate Real-Time Dispatch Tracking and Management:** Develop a synchronized dispatch tracking feature that allows residents to monitor the status of their reports (e.g., Pending, Dispatched, Resolved) while providing dispatchers with a centralized web dashboard interface to receive, assess, prioritize, and update incident statuses in real time.
4. **Provide Immediate Access to Emergency Contact Directories:** Integrate an offline-accessible emergency directory within the mobile application containing verified contact numbers for local responder agencies in Bacolod City, including police stations, fire stations, hospitals, and barangay offices.

Scope and Limitation

This study focuses on the design, development, and functional implementation of the AGAP-Link: Community Emergency Reporting and Incident Dispatch System, tailored specifically for the geographic and administrative boundaries of Bacolod City. The system encompasses two primary components: the citizen-facing AGAP-Link Mobile Application and the centralized dispatcher web dashboard. Developed natively for the Android operating system, the mobile application is designed for Bacolod City residents and features one-tap emergency category selection for structural fires, medical distress, and security threats. It also integrates automated capture and transmission of GPS coordinates, photo evidence attachment, a localized offline report queuing engine that automatically synchronizes data once network connectivity is restored, real-time dispatch status tracking, and an offline emergency hotline directory.

On the administrative side, the centralized dispatcher web dashboard is designed for authorized dispatchers and local emergency management personnel to receive, review, and organize incoming incident reports in real time. The dashboard displays incident categories, attached photographs, submission timestamps, and mapped GPS locations, enabling dispatchers to evaluate reports, update their operational status from pending to dispatched or resolved, and coordinate incident details directly with corresponding local response units, such as the Bureau of Fire Protection, emergency medical services, and local police stations.

Several operational and technical constraints boundary the scope of this study. Geographically, system deployment, incident logging, and directory data are strictly confined to the administrative boundaries of Bacolod City, meaning incident reports originating outside this jurisdiction are not handled by the system. In terms of connectivity, while the mobile application provides localized offline logging to queue reports during network dropouts, core operational functionalities—such as real-time alert transmission, live dispatch status synchronization, and dispatcher dashboard updates—strictly require an active cellular data or internet connection. Furthermore, accurate location logging is contingent upon the user's mobile device possessing functional and calibrated GPS hardware with active location permissions granted. The mobile application is developed exclusively for smartphones running the Android operating system, excluding legacy feature phones and devices running alternative mobile platforms. Finally, while AGAP-Link optimizes digital reporting, triage, and dispatch coordination workflows, physical responder transit times remain subject to external real-world factors beyond the software's control, including vehicle availability, personnel staffing levels, weather conditions, and urban traffic congestion.

Wireframe

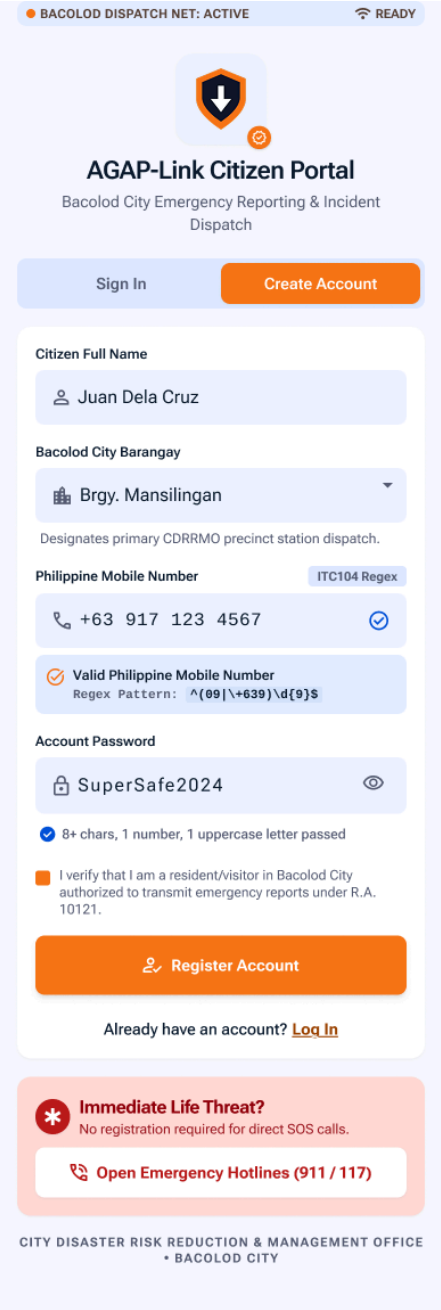


Figure 1. *Screen 1: Resident Registration & Login*

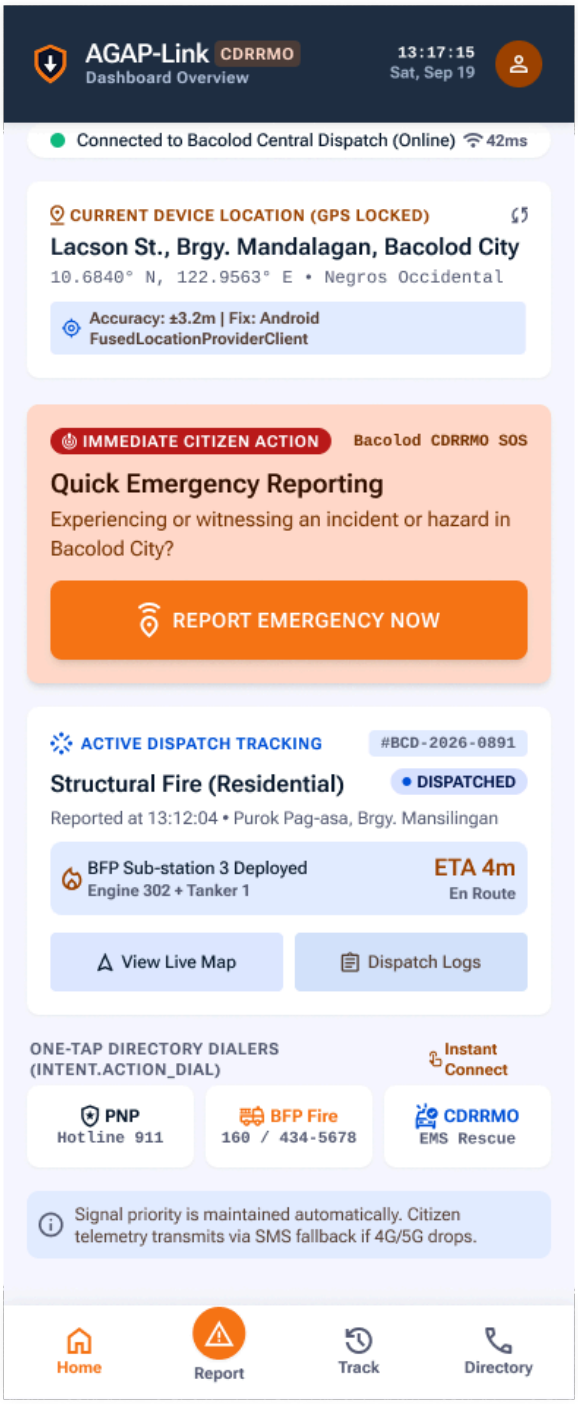


Figure 2. *Screen 2: Home Dashboard*

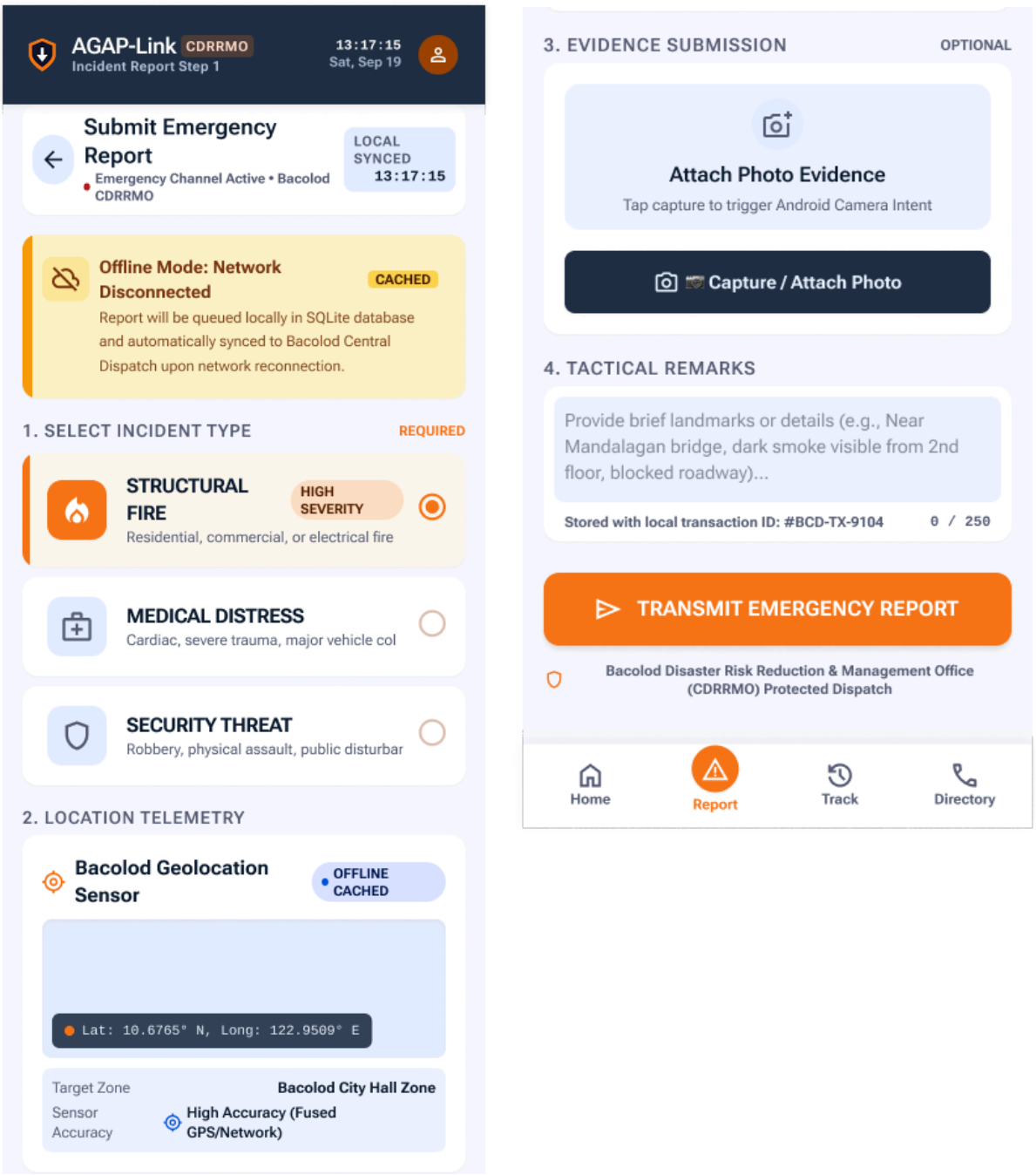


Figure 3. *Screen 3: One-Tap Report & Evidence Submission*

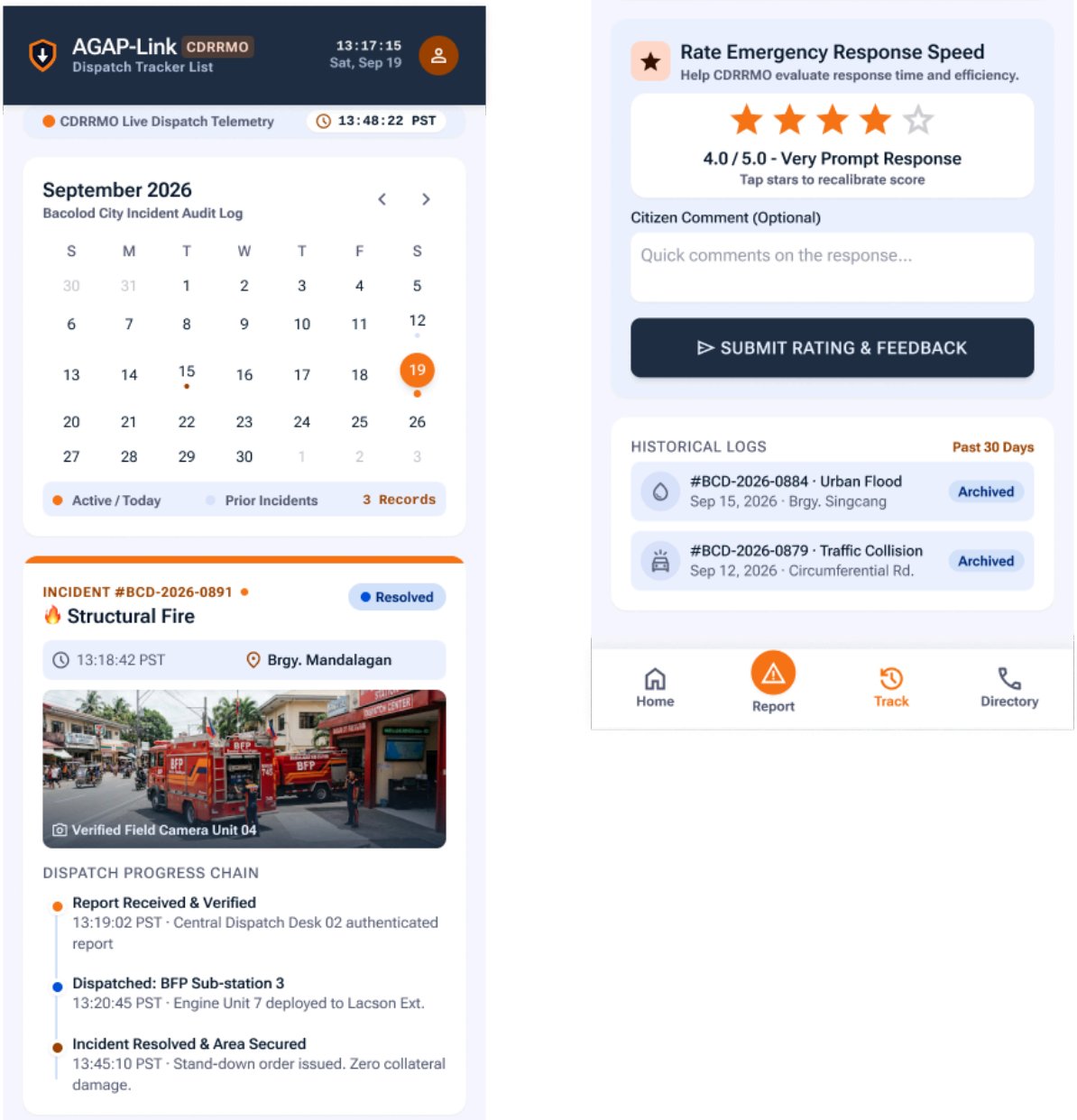


Figure 4. *Screen 4: Track & Incident History*

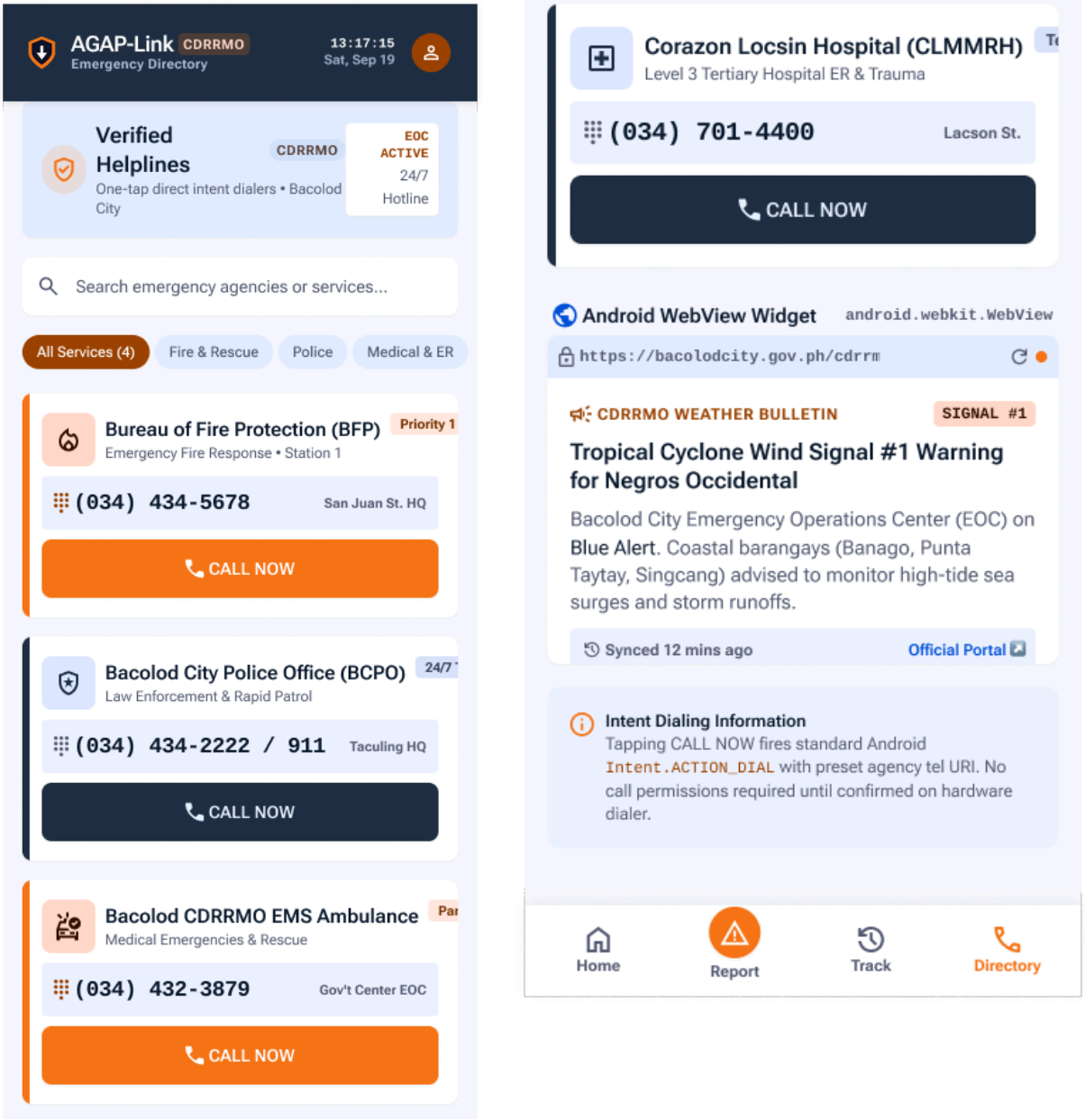


Figure 5. *Screen 5: Emergency Directory & Safety Advisories*

Use Case Diagram

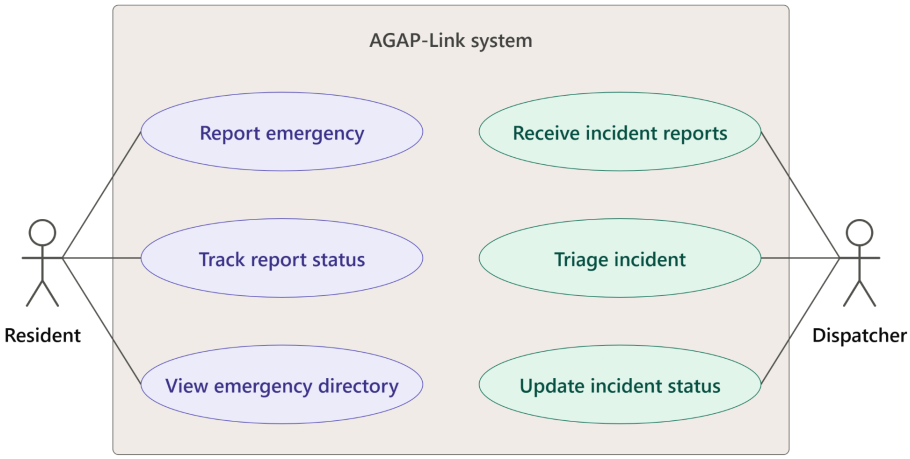


Figure 6. AGAP-Link Use Case Diagram

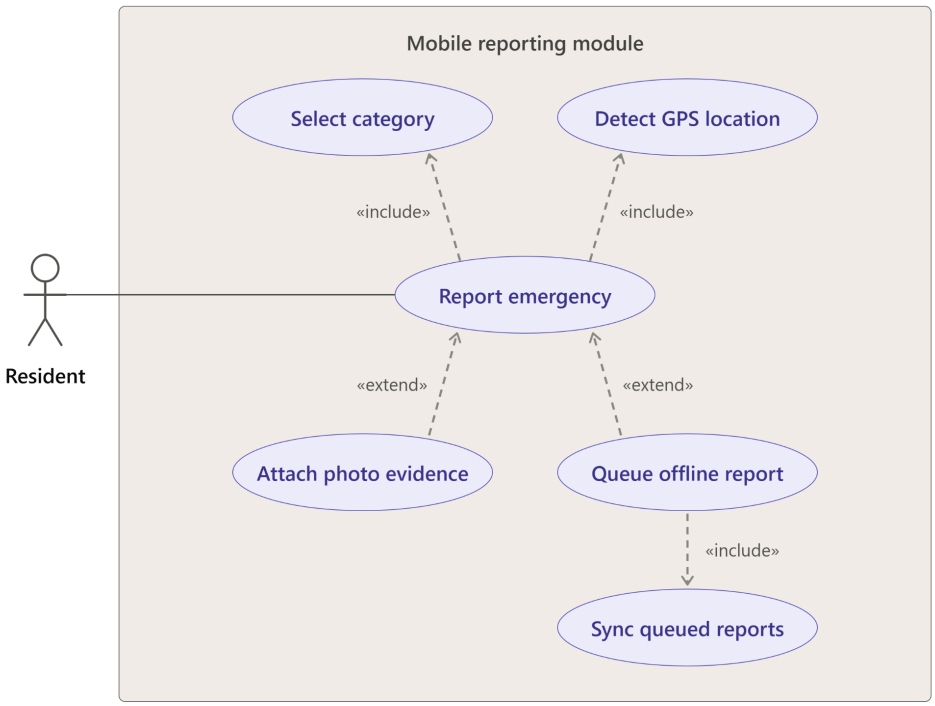


Figure 7. *Use Case Diagram of the AGAP-Link Mobile Reporting Module*

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